《数理金融》第六讲作业

1. 单时期市场 (S, D) 中, $S = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, $D = \begin{pmatrix} 1 & 2 & x \\ 2 & 3 & 5 \end{pmatrix}$, 其中 $x \geq 3$.

(1) 讨论当 x 满足什么条件时, 市场无套利, 并在无套利时求出状态价格向量集合.

解: 设 $D\psi = S \Rightarrow \begin{cases} \psi_1 + 2\psi_2 + x\psi_3 = 2 \\ 2\psi_1 + 3\psi_2 + 5\psi_3 = 3 \end{cases} \Rightarrow \begin{cases} \psi_1 = (3x-10)\psi_3 \\ \psi_2 = (5-2x)\psi_3 + 1 \end{cases}$ $\psi = \psi_3 \begin{pmatrix} 3x-10 \\ 5-2x \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

$\psi \gg 0$ 则 $\psi_3 > 0$, $\psi_1 > 0 \Rightarrow 3x-10 > 0 \Rightarrow x > \frac{10}{3}$, $\psi_2 > 0 \Rightarrow \psi_3 < \frac{1}{2x-5}$

(2) 对 $x = 3$ 时的市场特征构造一个套利组合.

集合为 $\{\psi \mid \psi = \psi_3 \begin{pmatrix} 3x-10 \\ 5-2x \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, 0 < \psi_3 < \frac{1}{2x-5}, x > \frac{10}{3}\}$

解: 套利组合 $\theta = (\theta_1, \theta_2)$, 设 $2\theta_1 + 3\theta_2 = 0 \Rightarrow \theta_1 = -\frac{3}{2}\theta_2$

$$\begin{cases} \theta_1 + 2\theta_2 = \frac{1}{2}\theta_2 \geq 0 \\ 2\theta_1 + 3\theta_2 = 0 \\ 3\theta_1 + 5\theta_2 = -\frac{9}{2}\theta_2 + \frac{10}{2}\theta_2 = \frac{1}{2}\theta_2 > 0 \end{cases}$$

则取 $\theta = (-\frac{3}{2}, 1)$ 为套利组合

(3) 该市场是否完全, 说明理由.

答: 不完全, 因为在无套利条件下 状态价格向量不唯一

(4) 对 $x = 4$ 时的市场, 对未定权益 $X = \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}$ 进行定价分析.

解: X 的价格 $h = X^T \psi$ $\psi = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \psi_3 + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2\psi_3 \\ 1-3\psi_3 \\ \psi_3 \end{pmatrix}$ $0 < \psi_3 < \frac{1}{3}$

则 $h = 8\psi_3 + 2 - 6\psi_3 + \psi_3 = 3\psi_3 + 2$, 其中 $0 < \psi_3 < \frac{1}{3}$

2. 单时期市场 (S, D) 中, $S = \begin{pmatrix} 2 \\ 3 \\ x \end{pmatrix}$, $D = \begin{pmatrix} 1 & 2 & y \\ 2 & 5 & 3 \\ 1 & 5/2 & 3/2 \end{pmatrix}$.

(1) 证明市场无套利的充要条件为 $x = 3/2, y > 2$.

无套利 $\Leftrightarrow D\psi = S$ 有解 $\psi \gg 0$ $\Leftrightarrow \begin{cases} \psi_1 + 2\psi_2 + y\psi_3 = 2 \cdots ① \\ 2\psi_1 + 5\psi_2 + 3\psi_3 = 3 \cdots ② \\ \psi_1 + \frac{5}{2}\psi_2 + \frac{3}{2}\psi_3 = x \cdots ③ \end{cases}$

$2\times③ - ②$ 得 $0 = 2x - 3 \Rightarrow x = \frac{3}{2}$

$② - ①\times 2$ 得 $\psi_2 + (3-2y)\psi_3 = -1$

$\Rightarrow \begin{cases} \psi_1 = (6-5y)\psi_3 + 4 > 0, \psi_3 > 0 \\ \psi_2 = (2y-3)\psi_3 - 1 > 0 \end{cases}$ $(5y-6)\psi_3 < 4$

$y > \frac{3}{2}, \psi_3 > \frac{1}{2y-3}$ $\psi_3 < \frac{4}{5y-6}$

$\frac{1}{2y-3} < \frac{4}{5y-6} \Rightarrow y > 2$

(2) 当 $x = 3/2, y = 3$ 时, 若益 $X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$ 可复制, 请给出 X_1, X_2, X_3 满足的条件.

$\exists \theta$ s.t. $D^T\theta = X \Rightarrow \begin{cases} \theta_1 + 2\theta_2 + \theta_3 = X_1 \\ 2\theta_1 + 5\theta_2 + \frac{3}{2}\theta_3 = X_2 \\ 3\theta_1 + 3\theta_2 + \frac{3}{2}\theta_3 = X_3 \end{cases} \Rightarrow \begin{cases} \theta_1 + 2(\theta_2 + \frac{1}{2}\theta_3) = X_1 \\ 2\theta_1 + 5(\theta_2 + \frac{1}{2}\theta_3) = X_2 \\ 3\theta_1 + 3(\theta_2 + \frac{1}{2}\theta_3) = X_3 \end{cases}$

消去 $\theta_1, \theta_2, \theta_3$, 得到 $3X_2 + X_3 = 9X_1$

$$\begin{pmatrix} 1 & 2 & x_1 \\ 2 & 5 & x_2 \\ 3 & 3 & x_3 \end{pmatrix}$$

$$a+b=\frac{1}{3}x_3$$

$$b=\frac{1}{3}x_3-a$$

$$2a+\frac{5}{3}x_3-5a=x_2$$

$$\frac{5}{3}x_3-x_2=3a$$

$$\underbrace{-\frac{1}{3}x_2+\frac{5}{9}x_3}_{a}$$

$$a+\frac{2}{3}x_3-2a$$

$$\frac{2}{3}x_3-\left(-\frac{1}{3}x_2+\frac{5}{9}x_3\right)=x_1$$

$$\frac{2}{3}x_3+\frac{1}{3}x_2-\frac{5}{9}x_3=x_1$$

$$6x_3+3x_2-5x_3=9x_1$$

$$X=\underbrace{\begin{pmatrix}1\\2\\3\end{pmatrix}}_{Y_1}a+\underbrace{\begin{pmatrix}2\\5\\3\end{pmatrix}}_{Y_2}b$$

线性相关

$$\det(Y_1\ Y_2\ Y_3)=0$$

$$x_1\begin{vmatrix}2&5\\3&3\end{vmatrix}-x_2\begin{vmatrix}1&2\\3&3\end{vmatrix}+x_3\begin{vmatrix}1&2\\2&5\end{vmatrix}$$

$$-9x_1-x_2(-3)+x_3$$

$$3x_2+x_3=9x_1$$